

Ayush Singhal

AI-ML Developer

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Summary

B.Tech Computer Science student with a strong interest in Artificial Intelligence and Machine Learning, seeking opportunities to apply and expand knowledge in building intelligent systems. Eager to gain hands-on experience in developing machine learning models, data analysis, and AI-driven applications while strengthening skills in Python, data science libraries, and modern AI technologies.

Technical Skills

- **Languages:** Python, C, C++, Java, Git, GitHub, Jupyter Notebook, Linux
- **Frontend:** Html, Css, javascript
- **Backend (JS, Python):** Flask, FastAPI
- **Databases:** MySQL; Machine Learning, Artificial Intelligence
- **Tools:** NumPy, Pandas, Matplotlib, Scikit-learn, SciPy, Seaborn, TensorFlow

Projects

AI Derma — AI Skin Analyzer | [GitHub](#) · *TensorFlow, OpenCV, Flask, React*

Feb 2026

- Developed an AI-powered skin analysis platform to detect 5+ common dermatological conditions using **TensorFlow CNN models** on webcam and uploaded images.
- Implemented real-time image preprocessing (resizing, normalization, noise reduction) to improve model accuracy by 12% and reduce inference time by 40%.
- Integrated a **Flask backend** with a React frontend, enabling seamless API-based communication and instant prediction feedback for 50+ concurrent users.
- Added **voice interaction (STT/TTS)** for hands-free usage, enhancing accessibility and engagement.
- Optimized the inference pipeline to achieve **low-latency predictions (<1s)** for real-time user interaction.

PredictX AI Portal | [GitHub](#) · [Live Demo](#) · *Scikit-learn, Python, Flask*

Jan 2026

- Built an AI-powered web platform enabling 100+ users to generate predictions without ML expertise.
- Implemented multiple machine learning models including **Gold Price Prediction, Credit Card Fraud Detection, Spam Classification, and Rainfall Prediction**.
- Designed modular Flask APIs to handle model inference, preprocessing, and result visualization for multiple prediction pipelines.
- Applied feature engineering and data preprocessing, improving model accuracy by up to 15% on real-world datasets.
- Delivered optimized prediction pipelines with **API latency <1s** and efficient handling of concurrent requests.

Experience

Technical Volunteer — AI ML Hub |

2025

- Assisted in organizing 5+ technical events and workshops; provided on-ground support including setup and troubleshooting for 200+ participants.
- Guided participants during activities, resolving 30+ technical issues and improving session efficiency.

Technical & Digital Volunteer — AITronics Hub |

2025

- Conducted online C++ workshops for 100+ students, improving average participant coding proficiency by measurable practice exercises.
- Edited and published 10+ short-form and long-form technical videos, reaching 1,000+ views across digital platforms.

Education

Jaypee Institute of Information Technology (JIIT), Noida

2024 – 2029

Integrated CSE (B.Tech + M.Tech)

SBV Senior Secondary School, Delhi

2024

Senior Secondary (PCM)

Certifications

- **Complete Data Science, Machine Learning, DL, NLP Bootcamp** — Udemy (Machine Learning, Deep Learning, NLP, hands-on projects, mathematical foundations)